XVA · QUANT SERIES

# XVA: The Price After the Price

CVA, DVA, FVA, KVA and MVA from first principles, with Python and QuantLib




Rates with Bip

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The AONIA and BBSW curves (from the AUD Swaps & Curves book), the SOFR, AUD/USD and cross-currency quotes (from the SOFR, FX & Cross-Currency Curves book), and the CDS spreads, swaption volatilities, AUD/USD volatility, funding spread and cost of capital in this book are illustrative, made up for teaching. They are not market data. Counterparty A, Counterparty W, the AA and BB clients and Our bank are fictional, and every simulated path is simulated.

Regulatory parameters (SA-CCR, risk weights, the reduced BA-CVA and the standard initial margin schedule) are checked against the Basel Framework as in force in 2026, and the crisis figure against the Basel Committee's 2011 press release; sources are listed at the end of each chapter. Capital and margin rules are simplified for teaching and change over time: consult the current Basel standards and APRA's prudential standards before relying on them. Nothing here is financial, accounting or regulatory advice.

Every number in the text, tables and figures is computed by the companion QuantLib Python notebooks (QuantLib 1.43), one per chapter, which reproduce each result.

This book is for education only. It is not financial, investment, legal or tax advice.

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# How to use this book

This book follows the fifteen video episodes of *XVA: The Price After the Price*, one chapter per episode, and goes a little further in each. It assumes you can price an interest-rate swap off a discount curve (the *AUD Swaps & Curves* book), know what collateral and cross-currency swaps are (*SOFR, FX & Cross-Currency Curves*), and have met Monte Carlo and option pricing (*Options from Zero*). Where it relies on one of those ideas, it says so.

The whole book follows one trade. We pay fixed on a five-year, AUD 100 million swap against six-month BBSW with Counterparty A, a fictional company rated BBB, with no collateral agreement, and hedge it with the opposite swap with a bank. Its risk-free value is zero. Chapter 1 adds up the bill that the risk-free price leaves out, and every later chapter takes one line of that bill apart.

- **Chapters 1–5: the ingredients.** What XVA is; default probabilities from credit default swaps; simulating the future of the interest-rate curve with Hull-White; exposure (EE and PFE); and the observation that ties the book together: expected exposure at a reset date *is* a swaption price.
- **Chapters 6–8: credit.** CVA by hand, by simulation and as a strip of swaptions; netting and how to split CVA between trades; and DVA, our own default.
- **Chapters 9–12: collateral and costs.** Variation margin and the margin period of risk; funding (FVA) and the debate it caused; regulatory capital (KVA) worked through Basel's formulas; and initial margin (MVA).
- **Chapters 13–15: putting it together.** Wrong-way risk; the full bill for an AUD/USD cross-currency swap; and how banks compute all of this at scale.

Each chapter ends with the chapter's one idea to remember, which the videos call *Bip's rule*, and a question to try before the next chapter; the answer opens the next chapter. The appendix has the final checkpoint with worked answers, and a glossary.

## Numbers you can reproduce

Every number in the text, tables and figures is computed by a companion Python notebook, one per chapter; nothing is typed in by hand. Simulations use fixed random seeds, so you will get the same numbers. Where the book works something by hand, or by a formula, the notebook checks it against a simulation or against QuantLib: CVA

three ways in Chapter 6, swaption prices against QuantLib's Jamshidian engine in Chapter 5, a regression estimate against QuantLib's finite-difference engine in Chapter 15.

The notebooks are free to download from github.com/RatesWithBip/curriculum, in the `xva-the-price-after-the-price` folder, and each has an *Open in Colab* button. They run as they are in Google Colab: each installs QuantLib and writes its own data files. The XVA toolkit they share, a few hundred lines of Python in `xva.py`, is pasted into every notebook.

## About the data

Every market input is illustrative, made up for teaching: the AONIA and BBSW curves of the first book, the SOFR and AUD/USD market of the second, and the CDS spreads, swaption volatilities, exchange-rate volatility, funding spread and cost of capital used here. Credit spreads and option quotes are licensed data, and the ideas don't depend on the exact levels. Every counterparty is fictional.

The regulatory numbers are real: risk weights, SA-CCR parameters, the basic approach to CVA capital and the standard initial margin schedule follow the Basel Framework as in force in 2026, and each chapter lists its sources. The rules are simplified for teaching, and they change.

## Free, and staying free

The videos, the notebooks and this book are free. If they help you, you can support the channel with a coffee at ko-fi.com/rateswithbip.

This book is for education only. It is not financial, accounting or regulatory advice.

CHAPTER 1 · VIDEO EPISODE 1

# The Price After the Price

*The risk-free price of a swap is only the first line of the bill. The rest of the bill is XVA.*

## Two clients, two rates

Ask a bank for a five-year Australian dollar swap and the rate depends on who you are. In this chapter's notebook the same swap costs an AA-rated client 4.4815% and a BB-rated client 4.5351%. Neither is mispriced. Both are the risk-free swap rate plus a bill for things the risk-free price assumes away.

The risk-free price of the earlier books projects cash flows off the BBSW curves and discounts them at the overnight rate, AONIA. That quietly assumes four things:

- nobody defaults, so every cash flow is paid;
- the bank can borrow whatever it needs at the overnight rate;
- holding the trade takes no shareholders' capital;
- nothing is locked up as initial margin.

Between two banks under a strong collateral agreement those assumptions are close to true. For a company that posts no collateral, none of them is.

## How the bill grew

Banks learned this in the global financial crisis. The Basel Committee reported that about two-thirds of the losses banks attributed to counterparty credit risk came from marking down trades with counterparties who *might* default (credit valuation adjustments), and only about a third from actual defaults. Basel III added a capital charge for that risk. Funding then became expensive, and banks began charging for it. From 2016, rules phased in requiring initial margin on derivatives that aren't centrally cleared; in Australia through APRA's CPS 226.

## The XVA family

The adjustments are known together as XVA, with X standing for whichever letter you need:

| LETTER | NAME | WHAT IT PRICES |
|---|---|---|
| CVA | credit valuation adjustment | the counterparty's default |
| DVA | debit valuation adjustment | our own default, seen from the other side |
| FVA | funding valuation adjustment | the cost of funding the trade and its hedge |
| ColVA | collateral valuation adjustment | collateral that earns something other than the overnight rate |
| KVA | capital valuation adjustment | the return shareholders want on regulatory capital |
| MVA | margin valuation adjustment | the cost of funding initial margin |

## The running trade

We pay fixed at the par rate of 4.4623% on AUD 100 million for five years against six-month BBSW, with Counterparty A, a fictional BBB-rated company. There is no collateral agreement. We hedge with the opposite swap with another bank, under a collateral agreement. The trade is worth zero on the risk-free curves.

Its value tomorrow is not zero. Simulating 40,000 futures for interest rates (Chapter 3), the average positive value peaks at about AUD 1.8 million, and in a bad case, one in forty, it reaches AUD 8.1 million. That positive part, what A could owe us, drives most of the bill.

## The bill

Each line is computed here with the simple form that a later chapter explains:

- **CVA.** A's five-year CDS spread of 120 basis points implies a 9.7% chance of default within the swap's life; with 40% recovery, the expected cost of A's default is AUD 62,756 (Chapter 6).
- **DVA.** When we owe A, our own default would cost A; our bank's 45 basis-point CDS spread gives AUD 18,821 in our favour (Chapter 8).
- **FVA.** When A owes us, we post collateral on the hedge but receive none from A, and borrow it at our funding spread of 45 basis points: AUD 25,649 (Chapter 10).
- **KVA.** Regulatory capital of about AUD 185,910 today, at a 10% cost of capital: AUD 93,576 over the life (a simplified version; Chapter 11 does it fully).

- **MVA.** Initial margin of AUD 4,000,000 on the hedge under the standard schedule, funded until maturity: AUD 33,023 (Chapter 12).

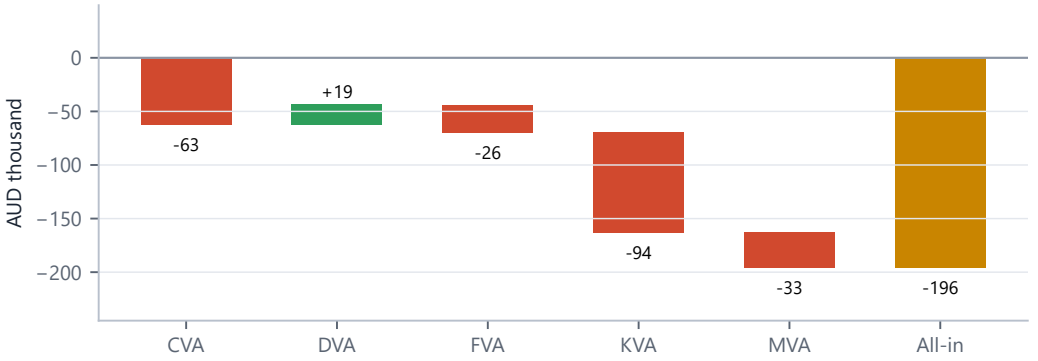



**Figure 1.1.** The running trade's bill in AUD thousand. Risk-free value zero; simplified formulas.

Altogether the trade is worth AUD 196,184 less than its risk-free value, or 4.38 basis points a year on the fixed rate: Counterparty A pays 4.5061% instead of 4.4623%. The biggest line is capital, 2.09 basis points, against 1.40 for CVA.

## Same swap, three clients

Funding, DVA and margin don't depend on who the client is; credit and capital do.

The same swap priced for three fictional clients. Risk weights: Basel standardised (AA 20%, BBB 75%, BB 100%).

| CLIENT | 5Y CDS (BP) | CVA (BP) | KVA (BP) | ALL XVA (BP) | FIXED RATE (%) |
|---|---|---|---|---|---|
| AA client | 40 | 0.48 | 0.56 | 1.92 | 4.4815 |
| Counterparty A (BBB) | 120 | 1.40 | 2.09 | 4.38 | 4.5061 |
| BB client | 320 | 3.60 | 2.79 | 7.28 | 4.5351 |

Inside a bank, an XVA desk owns this bill. It prices every adjustment before a trade is done, using the bank's whole portfolio with that client; it charges the trading desk, which passes the cost on in the client's rate; and it hedges what it can, because XVA moves every day.

**Bip's rule.** The risk-free price is only the first line of the bill.

**Try this.** If Counterparty A agreed to post collateral every day, which adjustments would shrink the most, and which could grow?

## Sources

Companion notebook: `s4e01_the_price_after_the_price.ipynb`. Every number in this chapter is computed by it.


1. Australian Prudential Regulation Authority, *APRA Prudential Standard CPS 226 Margining and risk mitigation for non-centrally cleared derivatives*. https://www.apra.gov.au/standards/cps-226
2. Bank for International Settlements, *Basel Committee press release on the Basel III counterparty credit risk (CVA) capital charge, June 2011*. https://www.bis.org/press/p110601.htm
3. Bank for International Settlements, *BCBS-IOSCO, Margin requirements for non-centrally cleared derivatives (2020 revision)*. https://www.bis.org/bcbs/publ/d499.pdf
4. Bank for International Settlements, *BCBS, The standardised approach for measuring counterparty credit risk exposures (bcbs279, 2014)*. https://www.bis.org/publ/bcbs279.htm

CHAPTER 2 · VIDEO EPISODE 2

# Default Odds from CDS Spreads

*Every XVA starts with the chance that someone defaults, and the market prices that chance in credit default swaps.*

## Last chapter's question

CVA would shrink the most, since A would rarely owe us much uncollateralised; FVA would mostly vanish, because A's collateral would fund the hedge. MVA could grow: under the margin rules both sides of a collateralised trade may have to post initial margin. Chapters 9 to 12 put numbers on each.

## A credit default swap

A credit default swap is insurance on a company's debt. The protection buyer pays a spread on the notional every quarter until the contract ends or the company defaults. If it defaults, the seller pays the loss: the notional times one minus the recovery rate. You don't need to own the debt to buy protection. Contracts are standardised: quarterly payment dates on the 20th of March, June, September and December, fixed coupons with an upfront payment, and a standard recovery assumption of 40% for senior unsecured debt.

What counts as a default is set by the contract's *credit events*: bankruptcy, failure to pay and, for some contracts, restructuring. A committee of dealers and investors rules whether an event happened, and an auction of the defaulted debt sets the recovery price.

## The credit triangle

If the company defaults at a constant rate $\lambda$ per year (the *hazard rate*), the seller expects to pay $\lambda(1-R)$ a year, and a fair spread must cover it:

$$s \approx \lambda\,(1-R) \quad \Rightarrow \quad \lambda \approx \frac{s}{1-R}, \qquad S(t) = e^{-\lambda t}.$$

For Counterparty A, 120 basis points over a 60% loss give $\lambda = 2.00\%$, a five-year survival probability of 90.48%, and a default probability of 9.52%. The full bootstrap below gives 9.72%: the triangle is close but not exact, because A's spreads rise with maturity.

## Bootstrapping the hazard curve

The bootstrap works like the curve builds of the first book. Solve the one-year segment's hazard rate so that the one-year contract reprices; hold it, solve the next segment from the two-year contract, and so on. QuantLib's `PiecewiseFlatHazardRate` does it from `SpreadCdsHelper`s.

Counterparty A's CDS curve, bootstrapped. Standard contracts mature on 20 December.

| CDS | SPREAD (BP) | TRIANGLE (%) | HAZARD IN SEGMENT (%) | SURVIVAL TO MATURITY (%) |
|---|---|---|---|---|
| 1Y | 60 | 1.00 | 1.01 | 98.76 |
| 2Y | 75 | 1.25 | 1.59 | 97.20 |
| 3Y | 90 | 1.50 | 2.13 | 95.15 |
| 5Y | 120 | 2.00 | 2.98 | 89.63 |
| 7Y | 135 | 2.25 | 3.12 | 84.21 |
| 10Y | 150 | 2.50 | 3.42 | 75.98 |

Each segment's hazard rate is above the triangle's average for that maturity because the spread curve slopes upward: from three to five years it is 2.98%. The check is the usual one. Reprice each of the 6 contracts on the finished curve; the largest difference from its quote is 1.3e-10 basis points.

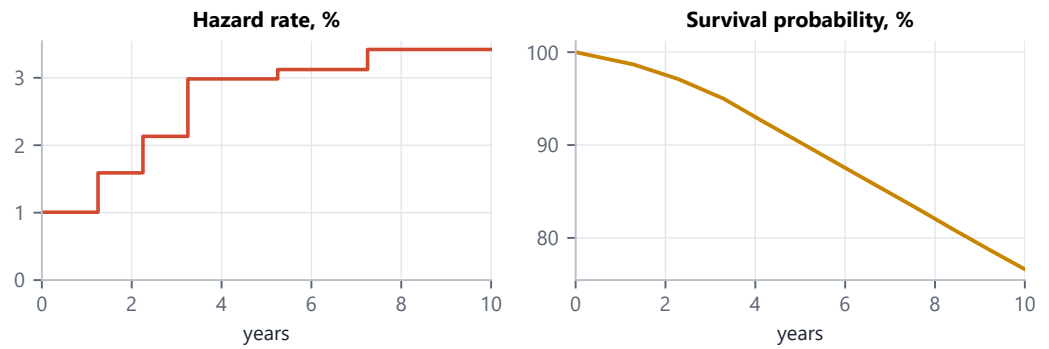



**Figure 2.1.** Counterparty A's hazard rate (a staircase, one step per quote) and the survival curve it implies.

## Default probabilities by year

XVA needs the chance of default in each slice of time, $\Delta PD_i = S(t_{i-1}) - S(t_i)$, seen from today. For A that is 1.00% in the first year rising to 2.67% in the tenth; 23.4% within ten years. Read these carefully: the chance of defaulting in year five seen from today is 2.73%, but given survival to year four (probability 93.0%) it is 2.94%. CVA uses the first.

The four fictional names of the book, each built the same way:

Default probabilities implied by illustrative CDS curves, recovery 40%.

| NAME | 5Y CDS (BP) | DEFAULT IN 1 YEAR (%) | DEFAULT IN 5 YEARS (%) |
|---|---|---|---|
| AA client | 40 | 0.33 | 3.33 |
| Our bank | 45 | 0.42 | 3.73 |
| Counterparty A | 120 | 1.00 | 9.72 |
| BB client | 320 | 2.97 | 24.19 |

## Upfronts and standard coupons

Contracts trade with a fixed coupon, 100 basis points for investment-grade names, and settle the difference upfront. A's fair spread is 119.6 basis points, so the buyer pays the extra once: about twenty basis points a year times a risky annuity of 4.58, an upfront of 0.90% of notional.

## Does recovery matter?

Recovery is a guess. Re-bootstrap from the same spreads with 25% recovery instead of 40%: the five-year default probability falls from 9.72% to 7.84%, because each default now costs more. But the CVA of the running trade moves by only 0.6%: the spread pins down loss times probability, and that product is what CVA uses.

## Prices, not forecasts

These are risk-neutral probabilities: investors demand a premium for bearing default risk, so spread-implied probabilities are usually higher than historical default rates. CVA is a price, so it uses the market's odds. In practice most companies have no CDS that trades, and the XVA desk builds a proxy curve from similar names by rating, sector

and region; think of Counterparty A's curve as a proxy for a BBB Australian industrial. Spreads move daily: 10 basis points wider everywhere takes A's five-year default probability to 10.48%.

The bootstrap in xva.py.

```
def credit_curve(spreads_bp, recovery, discount):
    """Bootstrap a flat-hazard credit curve from CDS par spreads {tenor: bp}.

    Standard CDS terms: quarterly premiums on CDS2015 dates, ACT/360, accrual paid on default.
    """
    helpers = [ql.SpreadCdsHelper(s / 1e4, ql.Period(t), 0, ql.WeekendsOnly(), ql.Quarterly, ql.Following,
                                  ql.DateGeneration.CDS2015, ql.Actual360(), recovery, discount)
               for t, s in spreads_bp.items()]
    curve = ql.PiecewiseFlatHazardRate(today, helpers, dc)
    curve.enableExtrapolation()
    curve.nodes()  # bootstrap now, so the helpers are linked and can report implied quotes
    return curve, helpers
```

**Bip's rule.** A credit spread is a default probability in disguise.

**Try this.** Double every CDS spread of Counterparty A. Does its five-year default probability double?

---

### Sources

Companion notebook: `s4e02_default_odds.ipynb`. Every number in this chapter is computed by it.

1. ISDA / IHS Markit, *ISDA CDS Standard Model and standard contract terms*. https://www.cdsmodel.com/
2. Bank for International Settlements, *Basel Framework MAR50, Credit valuation adjustment framework*. https://www.bis.org/basel_framework/chapter/MAR/50.htm

CHAPTER 3 · VIDEO EPISODE 3

# Simulating Tomorrow's Curve

*To know what a swap might be worth in two years, we need the whole curve in two years, thousands of times, consistent with today's prices.*

## Last chapter's question

Almost, but not quite: from 9.72% to 18.70%, a factor of 1.92. You can only default once, so survival compounds, and doubling the hazard rate less than doubles the chance of default.

## Why a model

CVA needs what the swap could be worth on every future date, on every path the world might take. That needs the whole AUD curve on each date. The futures can't be arbitrary: on average they must give back today's curve, and they must spread out by as much as the options market says rates will move.

## Hull-White

The Hull-White model (1990) describes the overnight rate $r$:

$$dr = \big(\theta(t) - a\,r\big)\,dt + \sigma\,dW.$$

Three dials, three sources. The drift $\theta(t)$ is chosen so the model reproduces today's AONIA curve exactly. The mean reversion $a$ pulls rates back towards that drift; we fix it at 3%, as many desks do. The volatility $\sigma$, in basis points because rates are normally distributed in this model, comes from swaption prices. Normal rather than lognormal is deliberate: rates move by similar amounts in basis points whatever their level, and can go below zero, and rate options are now usually quoted in normal (Bachelier) volatility.

## Calibration

Our trade lives five years, so we calibrate to the co-terminal swaptions that all end at year five, with QuantLib's `HullWhite` model and `JamshidianSwaptionEngine`:

Calibrating σ to illustrative normal volatilities.

| SWAPTION | MARKET VOL (BP) | MODEL VOL (BP) | ERROR (BP) |
|---|---|---|---|
| 1Y into 4Y | 90.0 | 89.2 | -0.80 |
| 2Y into 3Y | 90.0 | 89.2 | -0.79 |
| 3Y into 2Y | 89.0 | 89.3 | 0.26 |
| 4Y into 1Y | 88.0 | 89.3 | 1.27 |

The best single $\sigma$ is 94.0 basis points, within 1.3 basis points of vol on every quote. With constant $\sigma$ the model's co-terminal vols are almost flat whatever the mean reversion; fitting a sloping vol term structure needs a time-dependent $\sigma$ (QuantLib's `Gsr`).

## Exact simulation, and the martingale test

Write $r(t) = x(t) + \varphi(t)$, where $x$ is a mean-reverting Gaussian process starting at zero and $\varphi$ is fitted to the curve. Then $x(t)$ and its running integral are jointly normal with known means, variances and covariance, so we simulate them exactly, with no time-step error, and every future zero-coupon bond has a closed form:

$$P(t,T) = \frac{P(0,T)}{P(0,t)} \exp\Big(\tfrac{1}{2}\big[V(t,T) - V(0,T) + V(0,t)\big] - B(t,T)\,x(t)\Big),$$

with $B(t,T) = (1 - e^{-a(T-t)})/a$ and $V$ the variance of the integral of $x$. The notebook checks this formula against QuantLib's `discountBond` to machine precision.

The most important check in the book: averaged over all paths, the discount factor $e^{-\int_0^t r}$ must equal today's discount factor. If it doesn't, nothing priced with the model can be trusted.

The martingale test on 40,000 paths.

| YEAR | TODAY'S DISCOUNT FACTOR | AVERAGE OVER PATHS | GAP (BP) | ± ONE SE (BP) |
|---|---|---|---|---|
| 1 | 0.961547 | 0.961546 | -0.000 | 0.001 |
| 2 | 0.923125 | 0.923125 | 0.004 | 0.011 |
| 3 | 0.885615 | 0.885616 | 0.013 | 0.037 |
| 4 | 0.848128 | 0.848131 | 0.041 | 0.086 |
| 5 | 0.812192 | 0.812196 | 0.055 | 0.165 |

## What the paths look like

The overnight rate starts at 3.85% and fans out around today's forward curve; five years out its standard deviation is 195 basis points. On every path the whole curve can be rebuilt. The three-year swap rate two years from now has a bell-shaped distribution centred near today's forward swap rate of 4.36%, with a standard deviation of 129 basis points.

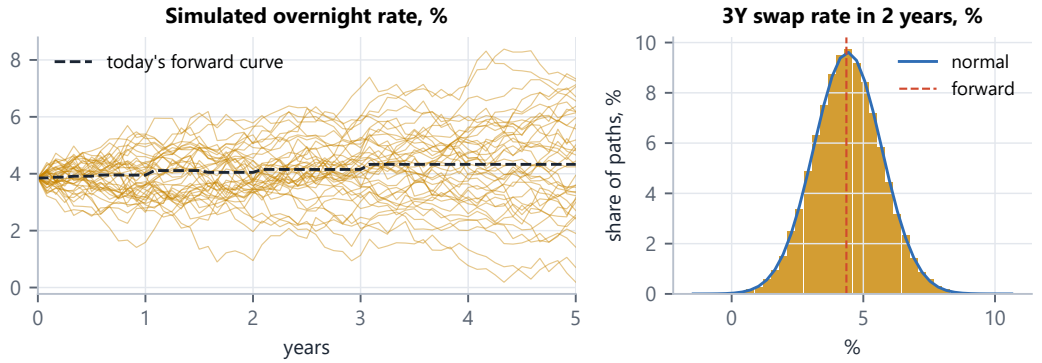



**Figure 3.1.** Left: 40 simulated paths of the overnight rate and today's forward curve. Right: the 3-year swap rate two years ahead across all paths.

## Mean reversion and one factor

A zero rate is linear in $x$, with sensitivity $B(t,T)/(T-t)$, so higher mean reversion damps long rates more than short ones:

Standard deviation of zero rates after one year, same σ.

| ONE YEAR FROM NOW | A = 3% (BP) | A = 10% (BP) |
|---|---|---|
| 1-year zero rate | 91 | 85 |
| 5-year zero rate | 86 | 70 |
| 10-year zero rate | 80 | 57 |

With one random driver, every rate moves together: the simulated one-year and ten-year rate changes have correlation 1.00. For one swap's exposure that matters little; for portfolios that hedge one part of the curve with another, desks use two or more factors. Nothing else in the book changes.

These paths are not a forecast. Their drift comes from today's curve and their spread from today's option prices: the risk-neutral world in which prices are fair.

**Bip's rule.** A good simulation gives back today's prices before it tells you anything new.

**Try this.** Can Hull-White produce negative interest rates? How often, five years out?

### Sources

Companion notebook: `s4e03_hull_white.ipynb`. Every number in this chapter is computed by it.

1. Oxford University Press, *Hull, J. and White, A. (1990), Pricing Interest-Rate-Derivative Securities, Review of Financial Studies 3(4)*. https://doi.org/10.1093/rfs/3.4.573

CHAPTER 4 · VIDEO EPISODE 4

# Exposure: EE and PFE

*Exposure is what you'd lose if your counterparty defaulted today, before recovery: the cost of replacing the trade.*

## Last chapter's question

Yes. Rates are normal in Hull-White, so some paths go negative: five years out the overnight rate averages 4.42% with a standard deviation of 195 basis points, and 1.1% of paths are below zero. With Australian rates around 4%, that is rare and harmless here.

## What exposure is

If Counterparty A defaults, our swap with A is gone, but we still want the position, so we replace it at the market price. If the swap was worth $V > 0$ to us, replacing it costs $V$. If it was worth something to A, we still owe that to A's administrators, and lose nothing. Exposure is $\max(V, 0)$.

Follow one path. On it rates rise: the overnight rate is 5.54% after two years, and since we pay a fixed rate set when rates were lower, the swap is worth AUD 4,833,000 to us.

One of 40,000 Hull-White paths.

| YEAR | OVERNIGHT RATE (%) | SWAP VALUE (AUD) | EXPOSURE (AUD) |
|---|---|---|---|
| 0 | 3.85 | 0 | 0 |
| 1 | 4.73 | 3,450,440 | 3,450,440 |
| 2 | 5.54 | 4,833,000 | 4,833,000 |
| 3 | 7.54 | 7,372,494 | 7,372,494 |
| 4 | 6.84 | 4,159,615 | 4,159,615 |

## EE, PFE and EPE

Across all paths, date by date:

- **EE(t)**, expected exposure: the average of $\max(V_t, 0)$; it peaks at AUD 1,810,762 about 2.5 years in.
- **PFE(t)**, potential future exposure: a high percentile, here 97.5%; it peaks at AUD 8,081,424, 4.5 times the EE peak.
- **EPE**, expected positive exposure: EE averaged over time, AUD 1,258,868 over five years.
- **Effective EPE**: the regulators' version, the average over the first year of the running maximum of EE, AUD 962,187.

To price, each path's exposure is discounted with that path's own rates: discounted, the EE peak is AUD 1,644,988. CVA uses the discounted profile; limits and capital usually use future dollars.

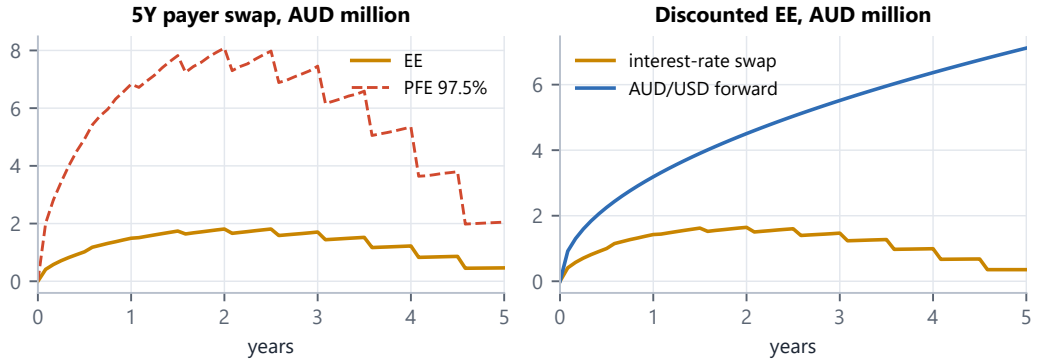



**Figure 4.1.** Left: EE and PFE of the 5-year payer swap (undiscounted). Right: discounted EE of the swap against a 5-year AUD/USD forward on the same notional.

## Why a swap's exposure humps

Two forces pull against each other. As time passes, rates have longer to move, so the swap's value wanders further from zero, pushing exposure up like $\sqrt{t}$. But fewer payments are left, so the same rate move matters less. Early on the first wins, later the second.

A paper rule of thumb captures both. Treat the swap's value at $t$ as normal with standard deviation $\sigma_N \sqrt{t}\, N A(t)$, where $\sigma_N$ is the swap rate's normal volatility and $A(t)$ the annuity of the remaining fixed payments. For an at-the-money swap the mean of the positive part is

$$\mathrm{EE}(t) \approx \frac{\sigma_N \sqrt{t}\, N\, A(t)}{\sqrt{2\pi}}.$$

The rule of thumb with the model's swaption vol against simulation.

| YEAR | RULE OF THUMB (AUD) | MONTE CARLO (AUD) | GAP (%) |
|---|---|---|---|
| 0.5 | 1,149,851 | 1,020,792 | 12.6 |
| 1 | 1,476,608 | 1,491,251 | -1.0 |
| 2 | 1,657,662 | 1,810,347 | -8.4 |
| 3 | 1,480,263 | 1,706,014 | -13.2 |
| 4 | 1,049,516 | 1,222,957 | -14.2 |

It lands within about 14% of the simulation. Most of the gap is drift: on an upward-sloping curve a payer swap's value tends to rise, which the rule ignores.

The profile also has teeth. Just before a coupon date the trade still contains that coupon's net payment; just after, it has been paid. Around the two-year coupon, EE drops by AUD 174,376 in 4 days, which is why every payment date belongs on the simulation grid.

## FX forwards don't hump

A five-year forward to buy AUD 100 million against US dollars pays nothing until the end, so nothing runs off: its exposure keeps growing with the exchange rate's uncertainty, from AUD 3.1 million after one year to AUD 7.1 million at maturity. Each point is a Black call on the forward, and the notebook checks it against QuantLib's `blackFormula`. Trades that exchange principal carry far more exposure than interest-rate swaps.

## Who uses which

EE drives prices. PFE drives permission: credit officers set a PFE limit per counterparty and check every new trade, added to the existing book, against it before it is done. For pricing we use risk-neutral paths; for limits, many banks also run real-world paths with drift and volatility from history. And PFE is not the worst case: one path in forty is worse by construction.

Notice that $\max(V, 0)$ is an option's payoff, and EE is its risk-neutral average. The next chapter takes that literally.

**Bip's rule.** Exposure is the part of a trade's future that the other side might not pay.

**Try this.** A ten-year swap on the same notional instead of five. Roughly how much bigger is its peak expected exposure?

## Sources

Companion notebook: `s4e04_exposure.ipynb`. Every number in this chapter is computed by it.

1. Bank for International Settlements, *Basel Framework CRE53, Internal models method for counterparty credit risk*. https://www.bis.org/basel_framework/chapter/CRE/53.htm

CHAPTER 5 · VIDEO EPISODE 5

# Expected Exposure Is a Swaption

*At a reset date, the positive part of a swap's value is a payer swaption's payoff. So expected exposure there is a swaption price.*

## Last chapter's question

About 2.3 times: AUD 3.8 million against AUD 1.6 million, peaking later, around 3.5 years in instead of 2. A longer life gives rates longer to move and leaves more payments at risk when they do.

## The identity

A payer swaption gives the right, at expiry, to enter a swap paying a fixed rate $K$. You exercise only if the swap is worth something, so it pays $\max(V, 0)$: exactly our exposure. At a reset date $t_k$ no coupon is running, the swap's value depends only on the model's state at $t_k$, and

$$\mathrm{EE}(t_k) = \mathbb{E}\big[D(0,t_k)\max(V_{t_k},0)\big] = P(0,t_k)\,\mathbb{E}^{t_k}\big[\max(V_{t_k},0)\big].$$

The second step changes to the measure in which the zero-coupon bond to $t_k$ is the unit of money (the $t_k$-forward measure, a change of numeraire from *Options from Zero*). Under it, Hull-White's state $x(t_k)$ is normal with mean

$$-\frac{\sigma^2}{a^2}\big(1-e^{-at_k}\big)+\frac{\sigma^2}{2a^2}\big(1-e^{-2at_k}\big)$$

and variance $\sigma^2(1-e^{-2at_k})/(2a)$, so the expectation is a one-dimensional integral: the price of a payer swaption expiring at $t_k$ on the rest of the swap, struck at our fixed rate of 4.4623%. We integrate on a dense grid of 8,001 points with the trapezoid rule; Gauss-Hermite quadrature converges badly on the kink of $\max(V,0)$ and was off by up to 0.8%.

## The strip

The running swap as a strip of payer swaptions, priced by integration, against Monte Carlo.

| EXPIRY | INTO | SWAPTION (AUD) | MONTE CARLO EE (AUD) | ± SE (AUD) |
|---|---|---|---|---|
| 0.51y | 4.5y | 1,093,219 | 1,089,762 | 5,200 |
| 1.01y | 4.0y | 1,399,500 | 1,402,666 | 6,269 |
| 1.51y | 3.5y | 1,495,500 | 1,500,837 | 6,584 |
| 2.01y | 3.0y | 1,489,680 | 1,486,723 | 6,343 |
| 2.51y | 2.5y | 1,383,471 | 1,387,299 | 5,784 |
| 3.01y | 2.0y | 1,218,391 | 1,226,858 | 4,936 |
| 3.51y | 1.5y | 967,622 | 970,301 | 3,953 |
| 4.02y | 1.0y | 671,161 | 669,327 | 2,754 |
| 4.51y | 0.5y | 355,094 | 354,184 | 1,459 |

At all 9 reset dates the two agree within 1.7 standard errors. A third check: on a single-curve version of the swap, QuantLib's own Hull-White pricer (`JamshidianSwaptionEngine`, which splits a swaption into options on zero-coupon bonds) agrees with the integration to AUD 0.23 on AUD 100 million.

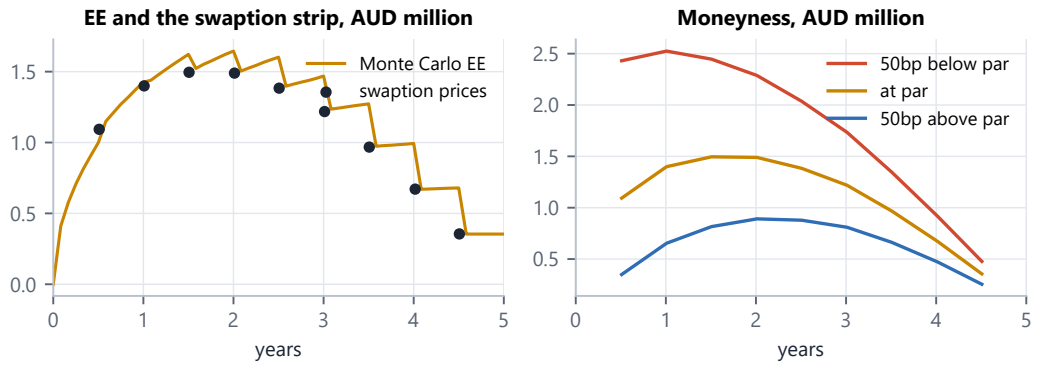



**Figure 5.1.** Left: swaption prices (dots) on the simulated EE curve, always at the bottom of a tooth. Right: EE at reset dates for fixed rates 50bp below, at and above par.

The hump of Chapter 4 now has an option explanation. The first swaption has little time to expiry but a long swap underneath (AUD 1,093,219); in the middle both are moderate and the swaption is worth most (AUD 1,495,500); the last has a long expiry but a short swap (AUD 355,094).

## From market quotes alone

We don't even need the model. Bachelier's formula with the market's normal volatility for each co-terminal swaption, today's forward swap rate $F$ and annuity $A$ of the remaining swap gives

$$\mathrm{EE} \approx NA\left[(F-K)\,\Phi(d) + \sigma\sqrt{t}\,\varphi(d)\right], \qquad d = \frac{F-K}{\sigma\sqrt{t}}.$$

Expected exposure straight from market quotes.

| EXPIRY | MARKET VOL (BP) | FORWARD (%) | BACHELIER EE (AUD) | MODEL EE (AUD) | GAP (%) |
|---|---|---|---|---|---|
| 1y | 90 | 4.543 | 1,411,262 | 1,399,500 | 0.8 |
| 2y | 90 | 4.604 | 1,501,795 | 1,489,680 | 0.8 |
| 3y | 89 | 4.667 | 1,216,042 | 1,218,391 | -0.2 |
| 4y | 88 | 4.667 | 663,310 | 671,161 | -1.2 |

The gaps, at most 1.2%, are the model's one-volatility fit of Chapter 3.

## Moneyness

Exposure behaves like any option. Agree to pay 50 basis points less than par and the swap is worth AUD 2.2 million to us today; the swaptions are in the money and peak EE rises to AUD 2.5 million. Pay 50 basis points more and it falls to AUD 0.9 million. Old trades that have moved in your favour carry the most exposure. Which swaption depends on direction: a payer's exposure is a strip of payer swaptions, a receiver's a strip of receiver swaptions, and the counterparty's exposure to us is the mirror image.

## Where the shortcut stops

Between reset dates a BBSW fixing has already been set on the path, so the value depends on two moments of the path: option-like, but not a plain swaption. And portfolios break it. Add trade B of Chapter 7, a two-year swap where we receive fixed at 4.16%. One year out, A alone has EE AUD 1,399,500 and B alone AUD 305,794, but under netting the exposure is an option on the *sum*, AUD 1,007,946, 41% less. That is why XVA engines simulate, and use results like the swaption strip to check the simulation.

Expected exposure by integration, in xva.py. It takes a list of swaps, so it nets them on common reset dates.

```
QUAD_Z = np.linspace(-9, 9, 8001)
QUAD_W = np.exp(-QUAD_Z**2 / 2) / math.sqrt(2 * math.pi) * (QUAD_Z[1] - QUAD_Z[0])


def ee_at_reset(hw, legs_list, t):
    """Discounted EE of a set of swaps at a date where none has a running fixing (e.g. a com-
mon reset date):
    P(0,t) x E[max(sum of values, 0)], with x(t) normal under the t-forward measure.
Trapezoid rule on a dense
    grid (Gauss-Hermite converges badly on the kink of max)."""
    m, s = hw.forward_measure_x(t)
    x = m + s * QUAD_Z
    v = sum(l.value_at(hw, t, x) for l in legs_list)
    return float(hw.P0(t) * np.dot(QUAD_W, np.maximum(v, 0)))
```

**Bip's rule.** Exposure is an option: price it like one, and check the simulation against it.

**Try this.** Double the rate volatility. What happens to the swap's expected exposure?

---

Companion notebook: `s4e05_exposure_swaption.ipynb`. Every number in this chapter is computed by it.

CHAPTER 6 · VIDEO EPISODE 6

# CVA: Pricing the Option to Default

*Counterparty A can walk away exactly when it owes us money. CVA is the price of that option.*

## Last chapter's question

It almost doubles: the peak goes from AUD 1,644,988 to AUD 3,118,761, 1.9 times. Each EE point is an at-the-money swaption, and an at-the-money option in a normal model is nearly proportional to volatility.

## The formula

If A defaults at time $t$, we lose the loss given default times the positive part of the swap's value. Averaging over when A defaults and what the swap is then worth, and slicing the trade's life into intervals:

$$\mathrm{CVA} = \mathrm{LGD} \sum_i \mathrm{EE}(t_i)\, \Delta PD_i, \qquad \Delta PD_i = S(t_{i-1}) - S(t_i),$$

with EE discounted (Chapter 4) and default probabilities from the CDS curve (Chapter 2). Three assumptions hide inside: recovery is the market's standard 40%, so LGD is 60%; the probabilities are risk-neutral, implied from CDS prices; and default is independent of the swap's value. Chapter 13 drops the last one.

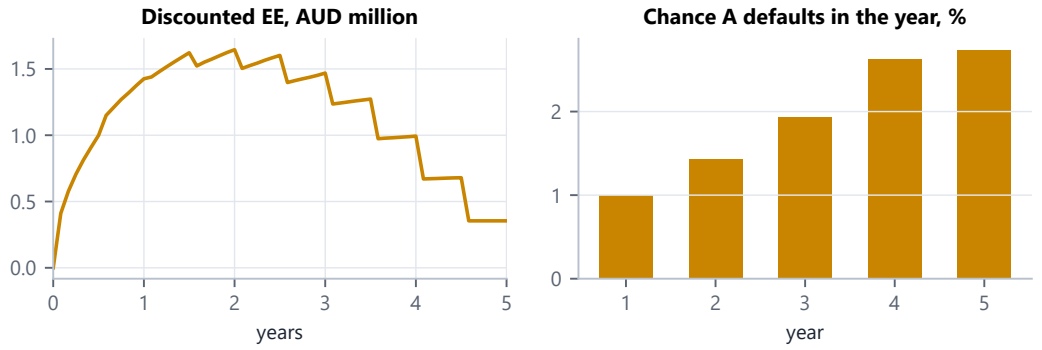



**Figure 6.1.** The two ingredients: the discounted EE profile of the running swap, and Counterparty A's chance of default in each year.

## CVA by hand

With one bucket per year and each year's average EE, the formula fits on a page:

CVA in yearly buckets. Total: AUD 62,823.

| YEAR | AVERAGE EE (AUD) | CHANCE A DEFAULTS (%) | × LGD = CVA (AUD) |
|---|---|---|---|
| Year 1 | 1,013,781 | 1.00 | 6,091 |
| Year 2 | 1,559,311 | 1.43 | 13,355 |
| Year 3 | 1,493,231 | 1.93 | 17,331 |
| Year 4 | 1,118,866 | 2.62 | 17,604 |
| Year 5 | 514,604 | 2.73 | 8,441 |

Averaging matters: exposure builds up before each coupon and drops when it is paid, so one date per year can land on a peak.

## CVA by simulation, and a third way

On a monthly grid, path by path, CVA is AUD 62,756 ± AUD 201, or 6.3 basis points of notional; the paper version was within 0.1%.

Chapter 5 gives a third route. At reset dates each EE point is a swaption price, so CVA on those dates is a strip of swaptions weighted by default probabilities: the Sorensen-Bollier formula (1994). On the same semi-annual dates it gives AUD 51,126 against a Monte Carlo AUD 51,201 ± AUD 159. Both are below the monthly answer because reset dates sit just after each coupon, at the bottom of every tooth: use a fine grid.

## CVA as a spread

Desks build CVA into the rate: divide by the notional times the annuity (4.48). That is 1.40 basis points a year, so A pays 4.4763% instead of the risk-free 4.4623%.

## What changes it

**Direction.** Flip to receiving fixed and CVA falls from AUD 62,756 to AUD 48,762. On an upward-sloping curve a payer pays more than it receives early and gets it back later, so its expected value drifts up (to about AUD 404,000) and more of its future is exposure.

**Credit quality.** Same exposure, three counterparties; CVA moves almost in proportion to the spread, the credit triangle at work:

Same swap, three fictional counterparties.

| COUNTERPARTY | 5Y CDS (BP) | CVA (AUD) | BP A YEAR |
|---|---|---|---|
| AA | 40 | 21,369 | 0.48 |
| BBB (Counterparty A) | 120 | 62,756 | 1.40 |
| BB | 320 | 161,378 | 3.60 |

**Credit spreads.** Widen every point of A's CDS curve by one basis point and CVA rises AUD 533: the CS01. A CVA desk hedges it by buying protection on A; about AUD 1.2 million of five-year CDS moves by the same amount.

**Interest rates.** Shift every AONIA quote up 10 basis points: the payer swap gains AUD 437,725 and CVA rises AUD 6,210 (9.9%). A CVA desk hedges rates too.

CVA is not a reserve for expected defaults set from history. It is a market price, revalued daily from spreads, and it moves profit and loss whether or not A ever defaults: in the crisis, those daily swings caused most of the counterparty losses.

Default buckets and CVA, in xva.py.

```
def default_buckets(credit, times):
    """Probability of default in each interval (times[i-1], times[i]], with times[0] = 0."""
    s = survival(credit, times)
    return np.concatenate([[0.0], s[:-1] - s[1:]])


def cva(ee, times, credit, lgd=LGD_STANDARD):
    """Unilateral CVA = LGD x sum of discounted EE x probability of default in the bucket (EE
at bucket end)."""
    return lgd * float(np.dot(ee, default_buckets(credit, times)))
```

**Bip's rule.** CVA is an exposure profile times a default curve.

**Try this.** Add a two-year swap with A where we receive fixed. Is the CVA of the two together more or less than the sum of their CVAs?

### Sources

Companion notebook: `s4e06_cva.ipynb`. Every number in this chapter is computed by it.

1. Bank for International Settlements, *Basel Committee press release on the Basel III counterparty credit risk (CVA) capital charge, June 2011*. https://www.bis.org/press/p110601.htm
2. ISDA / IHS Markit, *ISDA CDS Standard Model*. https://www.cdsmodel.com/
3. CFA Institute / Taylor & Francis, *Sorensen, E. H. and Bollier, T. F. (1994), Pricing Swap Default Risk, Financial Analysts Journal 50(3)*. https://doi.org/10.2469/faj.v50.n3.23

CHAPTER 7 · VIDEO EPISODE 7

# Netting, and Who Pays for the CVA

*Exposure to a netting set is an option on the sum of the trades, not a sum of options. So CVA doesn't add up, and allocating it is a choice.*

## Last chapter's question

Less. On their own the two trades carry AUD 62,945 and AUD 3,881 of CVA, AUD 66,827 in total; netted, AUD 58,424.

## Close-out netting

Banks trade with each counterparty under a master agreement, usually the ISDA Master Agreement. If one side defaults, every trade is terminated, valued and added into a single net amount. Netting only works if courts enforce it in insolvency; in Australia that protection comes from the Payment Systems and Netting Act 1998. Trade B is a two-year swap where we receive fixed at its par rate of 4.16%, quarterly against three-month BBSW. Both trades are simulated on one set of paths, on a grid that holds both trades' dates (so trade A's numbers differ from Chapter 6's by Monte Carlo noise only).

## Why netting helps

$$\max(V_A + V_B,\ 0)\ \leq\ \max(V_A, 0) + \max(V_B, 0)$$

path by path. When one trade is worth something to us and the other to A, they cancel. Here they point in opposite directions, and one year out their values have correlation -1.00: netted exposure is AUD 0.9 million against AUD 1.8 million separately.

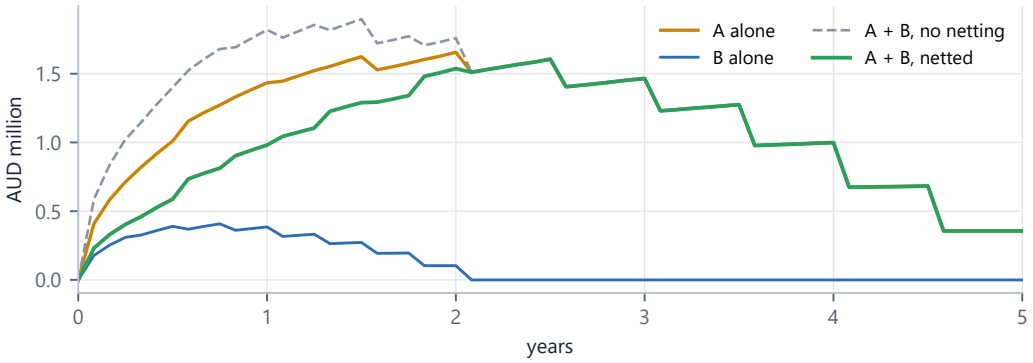



**Figure 7.1.** Discounted EE of each trade alone, of both without netting, and of the netting set.

The saving is 13% of CVA. Netting frees credit limits too: the netting set's peak PFE is AUD 8.0 million, below A's own AUD 8.1 million and far below the AUD 9.3 million you'd get by adding the two trades' PFEs.

## Who pays?

The XVA desk charges each trading desk for the CVA its trades create. But CVA belongs to the netting set. Two standard answers:

**Incremental CVA**, the change in the set's CVA when a trade is added, is what a new trade really costs. It depends on order:

Incremental CVA: the same two trades, two orders.

| ORDER OF ARRIVAL | FIRST TRADE (AUD) | SECOND TRADE (AUD) | NETTING SET (AUD) |
|---|---|---|---|
| A first, then B | +62,945 | −4,521 | +58,424 |
| B first, then A | +3,881 | +54,543 | +58,424 |

Arriving after A, B *reduces* the set's CVA: its incremental CVA is minus AUD 4,521, a rebate to the client.

**Euler (marginal) allocation** gives each trade its own share of the netted exposure on the paths where the set is owed money:

$$\mathrm{CVA}_i = \mathrm{LGD} \sum_k \Delta PD_k \; e_{i,k},$$

$$e_{i,k} = \mathbb{E}\big[D_k \, V_{i,k} \, \mathbf{1}\{V_k > 0\}\big].$$

Because the set's value is the sum of the trades' values, the shares add up exactly (Euler's theorem on homogeneous functions, the same idea used to split value at risk), and they don't depend on order.

Three ways to split the CVA. Stand-alone adds up to too much; incremental depends on history; Euler adds up.

| TRADE | STAND-ALONE (AUD) | INCREMENTAL, ARRIVING LAST (AUD) | EULER (AUD) |
|---|---|---|---|
| A: pay fixed, 5 years | +62,945 | +54,543 | +62,939 |
| B: receive fixed, 2 years | +3,881 | −4,521 | −4,515 |
| Netting set | +66,827 | +50,022 | +58,424 |

Compare a trade in the same direction: had B been a two-year payer, its incremental CVA would be AUD 4,528, 99.6% of its stand-alone AUD 4,545. Netting only helps trades that offset what's already there.

In practice desks price new trades on incremental CVA and report CVA by desk with Euler shares. A trade has no CVA of its own; its share can be negative; and netting needs an enforceable agreement in the counterparty's jurisdiction.

**Bip's rule.** CVA belongs to the netting set; trades only borrow a share.

**Try this.** Our own bank's CDS spreads widen by 10 basis points. Does the value of our swap with A go up or down?

### Sources

Companion notebook: `s4e07_netting.ipynb`. Every number in this chapter is computed by it.

1. International Swaps and Derivatives Association, *ISDA Master Agreement (2002) and ISDA netting opinions*. https://www.isda.org/
2. Federal Register of Legislation, Australia, *Payment Systems and Netting Act 1998 (Cth)*. https://www.legislation.gov.au/C2004A00395

CHAPTER 8 · VIDEO EPISODE 8

# DVA: Profiting from Your Own Decline

*We can default too, and that option is worth something to us. Accountants count it; regulators don't.*

## Last chapter's question

Up, by AUD 4,429. Our own default became more likely, and on paper that made us richer.

## The mirror image

When we owe A money and default, A recovers only part of it. From our side that is a benefit, the debit valuation adjustment, priced on our *negative* exposure $\mathrm{NEE}(t) = \mathbb{E}[D(0,t)\max(-V_t, 0)]$ and our own default probabilities:

$$\mathrm{DVA} = \mathrm{LGD}_{\mathrm{us}} \sum_i \mathrm{NEE}(t_i)\,\Delta PD_i^{\mathrm{us}},$$

$$\text{value} = \text{risk-free} - \mathrm{CVA} + \mathrm{DVA}.$$

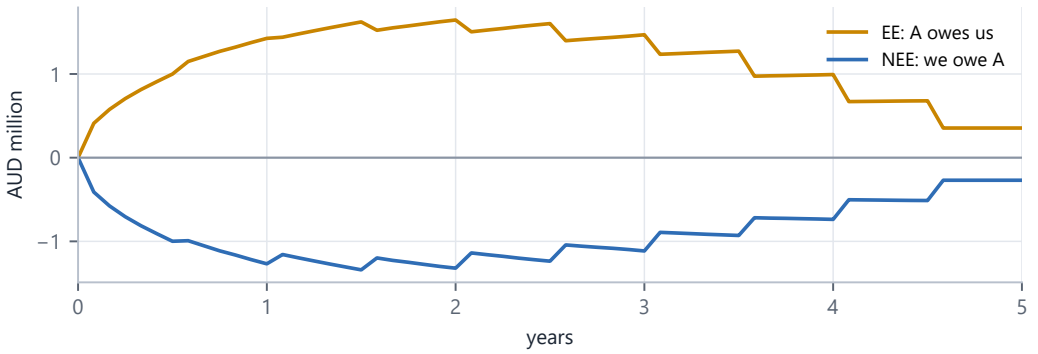



**Figure 8.1.** Expected exposure both ways: what A might owe us, and what we might owe A.

Our negative exposure peaks at AUD 1.3 million, a little below the positive side, because a payer swap tends to drift our way on an upward-sloping curve. With our bank's five-year CDS at 45 basis points (a 3.7% chance of default within five years), DVA is AUD 18,821, against CVA of AUD 62,756. The bilateral adjustment is AUD 43,935 against us, 0.98 basis points a year.

## Symmetry

Look from A's side, with the same curves. A's CVA (the cost of our default) is our DVA, and A's DVA is our CVA; the adjustment A makes is exactly the opposite of ours. Both sides agree on one price. With CVA alone each side would charge the other for credit, and no price would be fair to both.

Direction matters twice. If we received fixed instead, CVA would fall to AUD 48,762 and DVA rise to AUD 24,166.

## The paradox

Widen our own CDS curve by 10 basis points and DVA rises from AUD 18,821 to AUD 23,250: a gain of AUD 4,429, because our credit got worse. Push it further: if our bank were as risky as the BB client, our DVA would be AUD 126,049, more than the CVA we charge A, and under full bilateral pricing we would owe A a rate better than the risk-free one, worth AUD 63,293. The weaker the bank, the more generous its prices.

DVA can only be collected in full by defaulting. It can't be hedged directly, since nobody buys protection on you from you; banks sometimes sell protection on similar banks, which adds risk of its own. It does show up in real money through funding (Chapter 10).

## Accountants and regulators

Accounting fair value (IFRS 13) reflects the chance that a liability won't be paid, so DVA appears in reported profit and loss. Basel III removes gains from a bank's own credit from Common Equity Tier 1: a bank shouldn't strengthen its capital by becoming likelier to fail. Many banks charge clients full CVA but give little DVA credit in prices.

## First to default

Once either side defaults the trade is closed out, so A's default only matters if we haven't defaulted first. Weighting each side's default probabilities by the other side's survival:

First-to-default CVA and DVA.

| | EACH DEFAULT ALONE (AUD) | ONLY THE FIRST COUNTS (AUD) |
|---|---|---|
| CVA | 62,756 | 61,787 |
| DVA | 18,821 | 18,116 |
| CVA − DVA | 43,935 | 43,671 |

## Rates move DVA too

Raise AONIA 10 basis points: CVA rises AUD 6,210 and DVA falls AUD 1,973, so the bilateral adjustment worsens by AUD 8,184.

**Bip's rule.** DVA is real as a price and unreal as a profit.

**Try this.** If Counterparty A posted collateral every day with no threshold, would our exposure to A be zero?

### Sources

Companion notebook: `s4e08_dva.ipynb`. Every number in this chapter is computed by it.

1. Bank for International Settlements, *BCBS, Application of own credit risk adjustments to derivatives (consultative document, 2011) and Basel III definition of capital*. https://www.bis.org/publ/bcbs214.htm
2. IFRS Foundation, *IFRS 13 Fair Value Measurement*. https://www.ifrs.org/issued-standards/list-of-standards/ifrs-13-fair-value-measurement/

CHAPTER 9 · VIDEO EPISODE 9

# Collateral and the Margin Period of Risk

*Daily margin turns years of exposure into days of exposure. It never turns it into zero.*

## Last chapter's question

No. The peak falls to about 17% of what it was, but not to zero: when a counterparty defaults, collateral stops arriving, and it takes days to close out.

## The CSA

Collateral is governed by the credit support annex (CSA) to the master agreement. Its core is **variation margin**: every day, whichever side the market has moved against posts collateral equal to the trade's value. A **threshold** is exposure left uncovered by agreement; a **minimum transfer amount** avoids trivial calls; the CSA also sets eligible collateral, haircuts and the interest cash collateral earns. Between banks and large financial firms much of this is regulated: in Australia APRA's CPS 226 requires variation margin with a zero threshold, and a minimum transfer amount of at most AUD 750,000. Companies like Counterparty A are outside these rules.

## The margin period of risk

Follow a default slowly. On day zero A pays its last margin call. A call is missed, perhaps disputed, chased; default is declared; the trades are replaced at the new market price. Throughout, the swap keeps moving and we hold only day zero's collateral. Basel's floor for this gap, the margin period of risk (MPoR), is 10 business days for daily-margined bilateral trades and 20 for netting sets of more than 5,000 trades or with illiquid collateral or hard-to-replace derivatives, doubled after repeated disputes.

$$C(t) = V(t - \mathrm{MPoR}), \qquad E(t) = \max\big(V(t) - C(t),\ 0\big).$$

Collateralised exposure is about how far the swap can move in ten business days, not where it might be in five years. On one path, one year in, the swap is worth AUD 3.7 million and we hold AUD 2.9 million: we would lose only the gap, about AUD 786,500.

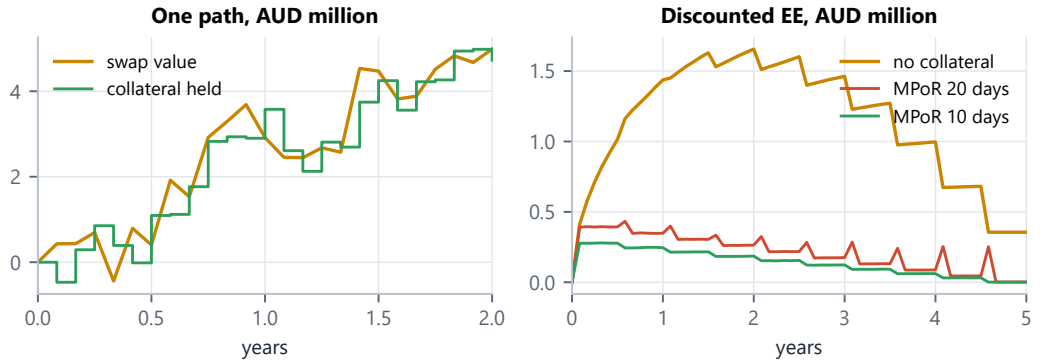



**Figure 9.1.** Left: one path's value and the collateral held (its value 10 business days earlier). Right: discounted EE without collateral and with daily margin.

## What it does to CVA

CVA with daily margin above a threshold, MPoR 10 business days.

| THRESHOLD | CVA (AUD) |
|---|---|
| zero | 6,319 |
| AUD 1 million | 24,145 |
| AUD 2 million | 39,412 |
| AUD 5 million | 58,919 |
| AUD 10 million | 62,873 |
| no collateral | 62,940 |

With daily margin and no threshold, CVA falls from AUD 62,940 to AUD 6,319, about 90% less. Double the margin period to 20 days and it rises to AUD 10,322, about 1.6 times as much: more than the square root of two that pure diffusion would give, because a longer gap is likelier to straddle a coupon payment. Thresholds give the benefit back quickly: one million takes CVA to AUD 24,145, five million to AUD 58,919. Clients want thresholds because posting cash daily costs liquidity and operations; the bank prices what is left uncovered.

## ColVA

Cash collateral normally earns AONIA, the rate the curves discount at, which is why collateralised trades are discounted at AONIA. Suppose the CSA pays AONIA minus 10 basis points. When we hold A's cash we pay 10 basis points less than it is worth; when we've posted, we earn 10 less. With collateral approximately equal to the trade's value,

$$\mathrm{ColVA} \approx 10\mathrm{bp} \times \sum_i \mathbb{E}\big[D(0,t_i)\,C(t_i)\big]\,\Delta t_i,$$

AUD 1,155 in our favour here, since a payer swap tends to drift our way. ColVA gets big when the CSA allows a choice of collateral currency or bonds: the poster picks the cheapest each day, and that choice is an option (the cross-currency story of the second book).

Collateralised is not risk-free: the close-out gap remains. A large threshold can undo the agreement. And collateral earns a rate, which is priced.

Collateral in xva.py.

```
def lag_index(grid, idx, days):
    """For each grid index in idx, the grid index closest to `days` calendar days earlier
(the margin period of risk)."""
    return np.array([int(np.argmin(np.abs(grid - (grid[k] - days / 365)))) for k in idx])


def collateral_held(V_lagged, threshold=0.0):
    """Two-way variation margin with a threshold, as last called: positive = we hold it, neg-
ative = we posted it."""
    return np.maximum(V_lagged - threshold, 0) - np.maximum(-V_lagged - threshold, 0)
```

**Bip's rule.** Collateral turns years of risk into days of risk.

**Try this.** If A posts collateral every day, do we still need to borrow money to run the hedge?

---

### Sources

Companion notebook: `s4e09_collateral.ipynb`. Every number in this chapter is computed by it.


1. Australian Prudential Regulation Authority, *APRA Prudential Standard CPS 226 and FAQ*. https://www.apra.gov.au/standards/cps-226
2. Bank for International Settlements, *Basel Framework CRE52, Standardised approach to counterparty credit risk (in force 1 Jan 2023)*. https://www.bis.org/basel_framework/chapter/CRE/52.htm

CHAPTER 10 · VIDEO EPISODE 10

# FVA and the Funding Debate

*Collateral you post but don't receive has to be borrowed, and borrowing costs money.*

## Last chapter's question

Mostly, no. When A owes us it posts collateral, and we pass it straight on to the bank on the other side of the hedge. Without collateral from A, we must find the cash ourselves.

## Following the cash

The client swap has no CSA; the hedge, with another bank, does. When rates rise, A owes us money on the client swap, the hedge owes the other bank the same amount, that bank calls for collateral, and we post it, borrowed at the overnight rate plus our funding spread of 45 basis points. When rates fall, the bank posts to us and that cash reduces our borrowing.

One path's funding bill.

| YEAR | CLIENT SWAP (AUD) | COLLATERAL POSTED (AUD) | FUNDING COST A YEAR (AUD) |
|---|---|---|---|
| 1 | 3,037,210 | 3,037,210 | 13,667 |
| 2 | 2,025,116 | 2,025,116 | 9,113 |
| 3 | 188,876 | 188,876 | 850 |
| 4 | -269,221 | 0 | 0 |

## FCA and FBA

Averaging over paths:

$$\mathrm{FCA} = s_f \sum_i \mathrm{EE}(t_i)\,\Delta t_i, \qquad \mathrm{FBA} = s_f \sum_i \mathrm{NEE}(t_i)\,\Delta t_i,$$

$$\mathrm{FVA} = \mathrm{FCA} - \mathrm{FBA}.$$

No default probability appears: it is exposure times a spread. For the running trade the funding cost is AUD 25,649 (0.57 basis points a year), the benefit AUD 20,397, the net AUD 5,252; for a receiver they swap places. Weighting each date by both sides' survival, since funding stops once either defaults, cuts FCA by 4.6% to AUD 24,472. FCA is linear in the funding spread: AUD 570 per basis point, so AUD 56,998 at 100 basis points, the kind of level bank funding reaches in a stress.

## The overlap with DVA

FBA is our negative exposure times our funding spread. DVA is our negative exposure times our default probability times loss given default. A bank's funding spread is largely its credit spread, and by the credit triangle (Chapter 2) a credit spread is default intensity times loss. So they measure the same thing: here FBA is AUD 20,397 against DVA of AUD 18,821. Funding spreads and CDS spreads aren't identical (funding also pays for liquidity, and deposits are cheap); this book sets both to 45 basis points so the overlap is plain.

The four lines for the running trade (negative = against us).

| LINE | AUD |
|---|---|
| CVA | −62,756 |
| DVA | +18,821 |
| FCA | −25,649 |
| FBA | +20,397 |

Counting all four gives −49,187 and counts our own credit twice. Most banks keep DVA and count only FCA (−69,584), or drop DVA and keep FBA (−68,008).

## The FVA debate

In 2012 John Hull and Alan White argued that funding costs should not affect a derivative's value: in a well-functioning market a value shouldn't depend on how its holder is funded, just as a bond's price doesn't depend on who buys it. Practitioners answered that the cash really is borrowed at a real spread, and a desk that ignores it loses money on every uncollateralised trade; Christoph Burgard and Mats Kjaer had already built funding into replication arguments (2011). Practice settled it, uneasily: in January 2014 JPMorgan reported a loss of about USD 1.5 billion on introducing FVA, and most large dealers followed. Many academics still say funding isn't part of value; every dealer's prices include it.

Inside a bank, treasury sets the internal funding curve and the XVA desk charges FVA at it. A client who signs a CSA removes most of the charge. And funding doesn't stop at variation margin: initial margin, posted on top and held until maturity, is funded too (Chapter 12).

**Bip's rule.** Collateral you post but don't receive has to be borrowed, and borrowing costs money.

**Try this.** Same swap, an AA-rated or a BB-rated client. Which uses more regulatory capital, and roughly how much more?

---

### Sources

Companion notebook: `s4e10_fva.ipynb`. Every number in this chapter is computed by it.


1. Infopro Digital, *Burgard, C. and Kjaer, M. (2011), Partial differential equation representations of derivatives with bilateral counterparty risk and funding costs, Journal of Credit Risk 7(3)*. https://doi.org/10.21314/JCR.2011.131
2. Risk.net, *Hull, J. and White, A. (2012), The FVA debate, Risk 25th anniversary edition, July 2012*. https://www-2.rotman.utoronto.ca/~hull/downloadablepublications/FVADebatePaper.pdf
3. JPMorgan Chase & Co., *JPMorgan Chase, fourth-quarter 2013 earnings (14 January 2014)*. https://www.jpmorganchase.com/ir/quarterly-earnings

CHAPTER 11 · VIDEO EPISODE 11

# Capital and KVA

*Regulators make banks hold capital against derivatives, and shareholders want a return on it. For uncollateralised corporate trades it is often the biggest line on the bill.*

## Last chapter's question

The BB client, by a lot: AUD 694,455 of capital today against AUD 240,965 for the AA client, about 2.9 times as much.

## Two capital charges

Capital is shareholders' money that absorbs losses before depositors do. A derivative with a counterparty attracts two charges: for the counterparty defaulting, and for CVA moving with credit spreads. In Australia APRA's APS 180 applies the Basel rules; this chapter follows the Basel Framework's standardised approaches.

## SA-CCR: exposure at default

The standardised approach for counterparty credit risk (Basel CRE52) uses fixed formulas, not our simulation. For one unmargined interest-rate swap:

$$SD = \frac{e^{-0.05S} - e^{-0.05E}}{0.05}, \qquad \text{add-on} = 0.5\% \times N \times SD \times MF,$$

$$\text{EAD} = 1.4\,(RC + \text{multiplier} \times \text{add-on}).$$

$S$ and $E$ are the start and end in years, so the supervisory duration is 4.43 and the adjusted notional AUD 442,643,170; the maturity factor $MF$ is 1 for unmargined trades with a year or more left; $RC = \max(V, 0)$ is zero today at par; the multiplier, 1 here, falls below 1 when the trade is worth less than zero (floored at 5%). The add-on is AUD 2,213,216 and

$$\text{EAD} = 1.4 \times \text{add-on} = \text{AUD } 3,098,502,$$

about twice our simulated expected exposure: a standard formula must be conservative for every bank. Alpha, the 1.4, is a supervisory buffer.

## Capital for default risk

Basel's standardised corporate risk weights (CRE20.43) are 20% for AA, 50% for A, 75% for BBB and 100% for BB. Risk-weighted assets are AUD 2,323,877, and minimum capital at 8% is AUD 185,910.

## Capital for CVA risk

The reduced basic approach (MAR50) for a single counterparty:

$$K = 0.65 \times \frac{RW}{1.4} \times M \times \mathrm{EAD} \times DF, \qquad DF = \frac{1 - e^{-0.05M}}{0.05M}.$$

The sector risk weight for an investment-grade industrial company is 3% (7% high yield), $M$ = 5.01 years, $DF$ = 0.88, giving AUD 191,389: about as much as the default charge. Total capital today: AUD 377,299.

Capital today for the same swap with three clients.

| CLIENT | DEFAULT RW (%) | CVA RW (%) | DEFAULT (AUD) | CVA RISK (AUD) | TOTAL (AUD) |
|---|---|---|---|---|---|
| AA client | 20 | 3 | 49,576 | 191,389 | 240,965 |
| Counterparty A (BBB) | 75 | 3 | 185,910 | 191,389 | 377,299 |
| BB client | 100 | 7 | 247,880 | 446,575 | 694,455 |

## KVA

Capital is held for the trade's life, so we project it. On every path and date, the replacement cost is the swap's value if positive, the add-on runs down with remaining maturity, and the multiplier applies when the value is negative; averaging over paths gives expected capital, which peaks around AUD 456,050.

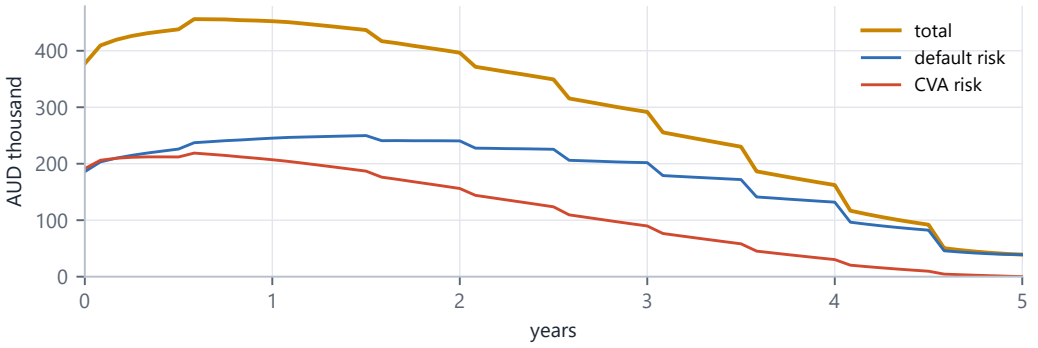



**Figure 11.1.** Expected regulatory capital over the trade's life, AUD thousand.

$$\mathrm{KVA} = h \sum_i \mathbb{E}[K(t_i)]\, P(0,t_i)\, \Delta t_i,$$

with an illustrative hurdle rate $h$ of 10%. KVA is AUD 137,359, or 3.07 basis points a year: 2.2 times the CVA of AUD 62,756. By client:

KVA by client, cost of capital 10%.

| CLIENT | KVA (AUD) | BP A YEAR |
|---|---|---|
| AA client | 75,937 | 1.70 |
| Counterparty A (BBB) | 137,359 | 3.07 |
| BB client | 236,747 | 5.28 |

Capital grows with credit risk, but in coarse regulatory buckets rather than in proportion to default probabilities.

## With a CSA

Under daily margin, SA-CCR's maturity factor drops to $1.5\sqrt{10/250}$ = 0.30, and replacement cost becomes the move since the last call. KVA falls to AUD 29,309, 79% less.

## Is KVA a value?

Sceptics say capital is a constraint, not a cash flow, and the hurdle rate is a choice, so KVA doesn't belong in fair value. Desks answer that capital is scarce and must earn a return. In practice most banks charge KVA in prices but don't book it as fair value. Some large banks may use internal models instead of SA-CCR, with supervisory approval, which typically gives numbers closer to simulated exposure.

SA-CCR exposure at default per path, in xva.py.

```
def sa_ccr_ead(V, addon, collateral=0.0, alpha=1.4):
    """SA-CCR exposure at default per path: alpha x (RC + multiplier x add-on), RC = max(V - C, 0), with the
    multiplier for negative value or excess collateral, min(1, 0.05 + 0.95 exp((V - C) / (2 x 0.95 x add-on)))."""
    vc = V - collateral
    if addon <= 0:
        return alpha * np.maximum(vc, 0)
    mult = np.minimum(1.0, 0.05 + 0.95 * np.exp(np.minimum(vc, 0) / (2 * 0.95 * addon)))
    return alpha * (np.maximum(vc, 0) + mult * addon)
```

**Bip's rule.** Capital is never free: someone has to be paid to hold it.

**Try this.** Initial margin is posted on top of variation margin. What risk does it cover that variation margin doesn't?

---

### Sources

Companion notebook: `s4e11_capital_kva.ipynb`. Every number in this chapter is computed by it.


1. Australian Prudential Regulation Authority, *APRA Prudential Standard APS 180 Capital Adequacy: Counterparty Credit Risk (1 January 2024)*. https://www.apra.gov.au/standards/aps-180
2. Bank for International Settlements, *Basel Framework CRE20.43, Table 10 (corporate risk weights)*. https://www.bis.org/basel_framework/chapter/CRE/20.htm
3. Bank for International Settlements, *Basel Framework CRE52 (SA-CCR)*. https://www.bis.org/basel_framework/chapter/CRE/52.htm
4. Bank for International Settlements, *Basel Framework MAR50 (CVA risk), reduced BA-CVA*. https://www.bis.org/basel_framework/chapter/MAR/50.htm

CHAPTER 12 · VIDEO EPISODE 12

# Initial Margin and MVA

*Variation margin covers today's value; initial margin covers the close-out gap. Funding it is the last line on the bill.*

## Last chapter's question

The gap. Variation margin keeps collateral up to date as of the last margin call; initial margin covers how far the value can move before close-out. With variation margin alone, our expected exposure still peaked at AUD 280,032 (Chapter 9).

## The rules

Initial margin is required on derivatives between banks and other large financial firms that aren't centrally cleared. Both sides post it, gross, into segregated accounts held by a third party, and margin received can't be reused. It is sized to cover a 99% move over ten days. The rules phased in from 2016, with the final phase in September 2022; in Australia they are part of APRA's CPS 226. Counterparty A, an ordinary company, is outside them, so for our trade initial margin hits the *hedge* with the other bank, which we must fund for its whole life.

## The standard schedule

Basel's schedule (MGN20.16) sets interest-rate margin by remaining maturity: 1% of notional under two years, 2% from two to five, 4% beyond five. Our hedge has just over five years left, so today it is AUD 4,000,000. Netting under the schedule is partial:

$$\mathrm{IM} = 0.4 \times \text{gross} + 0.6 \times \mathrm{NGR} \times \text{gross},$$

$$\mathrm{NGR} = \frac{\text{net replacement cost}}{\text{gross replacement cost}},$$

so even perfectly offsetting trades keep 40% of the gross amount.

## Model-based margin

Most margin is model-based, aiming at the 99th percentile of a ten-day move. Our stand-in is the simulated ten-business-day change in the swap's value: one month in, its 99th percentile is AUD 1,772,918, a move of about 40 basis points in the swap rate (ten-day standard deviation 15.5 basis points). The schedule asks for 2.3 times as much. The industry's shared model, ISDA's SIMM, works from sensitivities instead: deltas by tenor and vegas, times prescribed risk weights, combined with prescribed correlations, so both sides compute the same number. Its parameters are ISDA's licensed property; this book uses the idea, not the numbers.

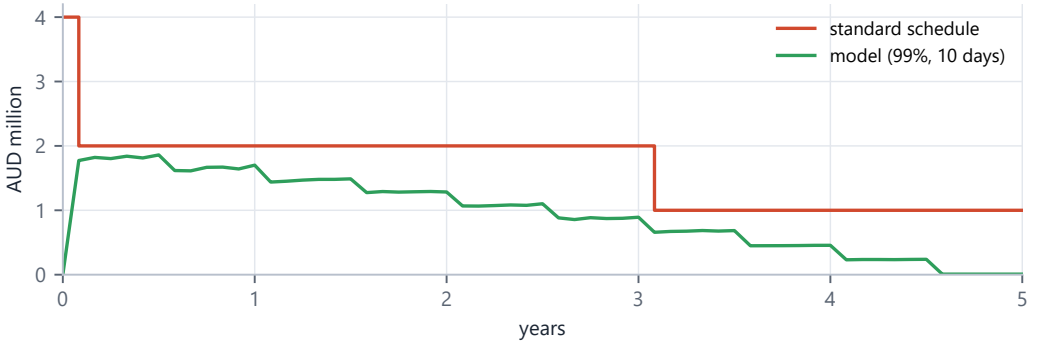



**Figure 12.1.** Initial margin on the hedge over time: the standard schedule steps down at five and two years to go; the model margin runs off smoothly.

## What margin does to CVA

If Counterparty A posted both kinds of margin, our exposure would be only the part of a ten-day move beyond the margin: rare by construction. CVA falls from AUD 62,858 with no margin, to AUD 6,330 with variation margin, to AUD 24 with both. Counterparty risk has been converted into a funding cost.

## MVA

$$\mathrm{MVA} = s_f \sum_i \mathbb{E}[\mathrm{IM}(t_i)]\, P(0,t_i)\, \Delta t_i,$$

FVA's cousin, with no benefit side because margin received is locked away. At our 45 basis-point funding spread:

| MARGIN | MVA | BP A YEAR |
|---|---|---|
| Standard schedule | AUD 33,023 | 0.74 |
| Model-based | AUD 20,111 | 0.45 |

The schedule costs 1.6 times as much, which is why banks invested so heavily in approved, agreed models.

## In practice

Many hedges between banks are centrally cleared: a central counterparty collects variation margin daily and initial margin from its own model, and members contribute to a default fund. Australian dollar swaps clear at ASX and at the global clearing houses, so MVA still applies, with the clearing house's model. For a real portfolio, projecting initial margin is one of the hardest problems in XVA: MVA needs the margin on every future path and date, and a sensitivity-based margin needs the portfolio's sensitivities there too (Chapter 15).

The margin rules swap one cost for another: less counterparty risk and capital, more funding and liquidity, with margin calls spiking exactly when cash is hardest to find.

**Bip's rule.** Margin turns credit risk into funding cost.

**Try this.** An Australian exporter earns US dollars and hedges by selling them forward to us for Australian dollars. Right-way or wrong-way risk?

### Sources

Companion notebook: `s4e12_initial_margin.ipynb`. Every number in this chapter is computed by it.


1. ASX, *ASX OTC clearing (ASX Clear (Futures))*. https://www.asx.com.au/
2. Bank for International Settlements, *BCBS-IOSCO, Margin requirements for non-centrally cleared derivatives (April 2020)*. https://www.bis.org/bcbs/publ/d499.htm
3. Bank for International Settlements, *Basel Framework MGN20.16 (standardised initial margin schedule) and MGN20.17 (net-to-gross)*. https://www.bis.org/basel_framework/chapter/MGN/20.htm

CHAPTER 13 · VIDEO EPISODE 13

# Wrong-Way Risk

*Every CVA so far assumed default tells us nothing about exposure. Wrong-way risk is when the counterparty is likeliest to fail exactly when it owes us most.*

## Last chapter's question

Right-way. The exporter owes us money when the Australian dollar falls, which is when its business is strongest: its US dollar revenues are worth more at home.

## The assumption

Multiplying expected exposure by default probabilities assumes the two are independent. **Wrong-way risk** is when default is likelier when exposure is high; **right-way risk** the reverse. *General* wrong-way risk works through the economy: a regional firm whose fortunes follow its home currency, or a commodity producer whose hedge pays us when prices collapse. *Specific* wrong-way risk is built into the deal, like a loan secured by the borrower's own shares; regulators treat it especially harshly.

## The example

Counterparty W is a fictional financial firm tied to the Australian economy, with a five-year CDS spread of 220 basis points. It buys AUD 100 million from us for US dollars in five years at the forward rate of 0.6802. We gain, and W owes us, when the Australian dollar falls, which is often when the Australian economy is in trouble. The exchange rate is lognormal at 10% volatility on the Series 2 curves, held fixed.

The forward's discounted exposure at $t$ is a Black put on the forward with volatility $\sigma\sqrt{t}$ ; Monte Carlo matches it within 1.8 standard errors, growing from AUD 3.2 million after one year to AUD 7.1 million near maturity. With W's 17.2% chance of default within five years, the independent CVA is AUD 519,337.

## Linking default to the exchange rate

A Gaussian copula mixes the exchange rate's standardised shock $Z$ over the five years with W's own randomness $\varepsilon$:

$$U = \Phi\big(-\rho\, Z + \sqrt{1-\rho^2}\,\varepsilon\big), \qquad \tau = S^{-1}(U).$$

Each path's default time still follows W's survival curve $S$, but with $\rho > 0$ the paths where the Australian dollar falls bring earlier defaults. With $\rho = 0$ the simulation must reproduce the independent formula, and it does: AUD 535,301 ± AUD 10,516.

CVA against correlation. Positive: wrong-way; negative: right-way.

| CORRELATION | CVA (AUD) | × INDEPENDENT |
|---:|---:|---:|
| -0.50 | 249,565 | 0.48 |
| -0.25 | 391,861 | 0.75 |
| 0.00 | 535,301 | 1.03 |
| 0.25 | 673,225 | 1.30 |
| 0.50 | 845,155 | 1.63 |
| 0.75 | 1,032,722 | 1.99 |

At a correlation of one half, CVA is AUD 845,155, 1.63 times the independent number; at minus one half (the exporter's case) it is AUD 249,565, less than half.

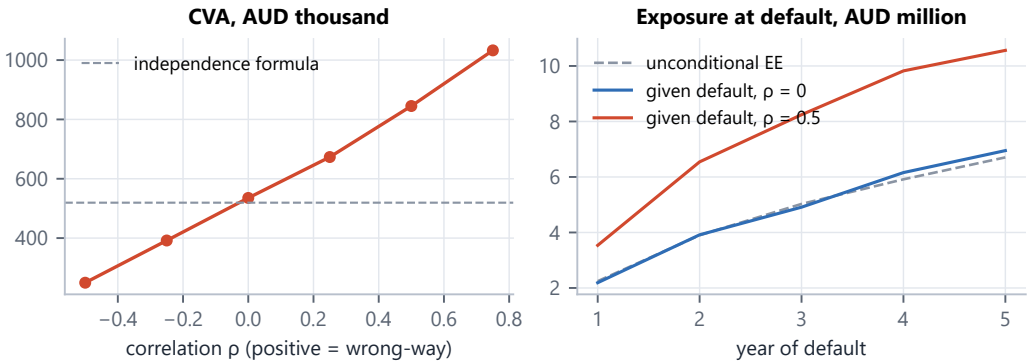



**Figure 13.1.** Left: CVA against correlation. Right: average exposure on the paths where W defaults in each year.

## Exposure at the moment of default

Look only at the paths where W defaults, and ask what it owed at that moment. With no correlation, it is the ordinary expected exposure. With a correlation of one half, in year three it is AUD 8.2 million against an average of AUD 5.0 million. The correct formula always uses exposure conditional on default,

$$\mathrm{CVA} = \mathrm{LGD} \sum_i E_i \, \Delta PD_i,$$

$$E_i = \mathbb{E}\big[D \max(V, 0) \mid \tau \in (t_{i-1}, t_i]\big],$$

and independence is the special case $E_i = \mathrm{EE}(t_i)$.

## Why it's hard

The correlation can't really be measured: defaults are rare and the link shows up mostly in crises. The copula is a modelling choice. A hedge means buying protection on W from someone not hit by the same crisis. So banks often answer wrong-way risk with limits and collateral rather than a price; regulators require banks to identify and stress it, and the 1.4 of Chapter 11 partly allows for it. Collateral helps, but crashes are fast, and collateral tied to the same risk loses value at the worst moment. Even our running interest-rate swap could be either: wrong-way with a heavily indebted floating-rate borrower, right-way if rates fall in the downturns when companies struggle. It depends on the counterparty's business, not the product.

**Bip's rule.** Ask not what they owe on average, but what they owe when they fail.

**Try this.** A cross-currency swap exchanges notionals at the start and end. Where does most of its exposure come from? And what if its notional resets every quarter?

---

**Sources**

Companion notebook: `s4e13_wrong_way_risk.ipynb`. Every number in this chapter is computed by it.

1. Bank for International Settlements, *Basel Framework CRE53, Internal models method: wrong-way risk*. https://www.bis.org/basel_framework/chapter/CRE/53.htm

CHAPTER 14 · VIDEO EPISODE 14

# XVA of a Cross-Currency Swap

*The whole bill for two five-year AUD/USD cross-currency swaps, one with a constant notional and one that resets every quarter.*

## Last chapter's question

From the exchange rate, through the principal swapped back at the end, as with a long-dated FX forward. A notional that resets every quarter wipes that principal risk clean four times a year.

## Two swaps

Both with Counterparty A, five years, quarterly from the spot date: we receive three-month BBSW plus a spread on AUD 100 million and pay SOFR on the US dollar equivalent, exchanging principals at the start and the end. With a **constant notional** the US dollar amount is fixed at the start (USD 71,000,000 at 0.7100). With a **mark-to-market** notional it resets each quarter to AUD 100 million times the exchange rate then, and the difference is paid in cash: the AUD/USD market standard of the second book. With interest rates held fixed, both have the same par spread of 15.5 basis points.

The model is the simplest that captures what matters: the Series 2 AUD and USD curves held fixed, and a lognormal exchange rate at 10% volatility. On every path and date, the Australian dollar flows still to come are worth the exchange rate times their present value, the US dollar flows their present value, and a resetting notional adds just the current quarter's repayment. The check: on today's exchange rate the path pricer reproduces the second book's valuation of the mark-to-market swap to USD 1.5e-08.

## Where the exposure is

We receive AUD and pay USD, so we are long Australian dollars: we are owed money when the AUD rises. With a constant notional that gain builds for five years; with resets, only one quarter's move is ever outstanding.

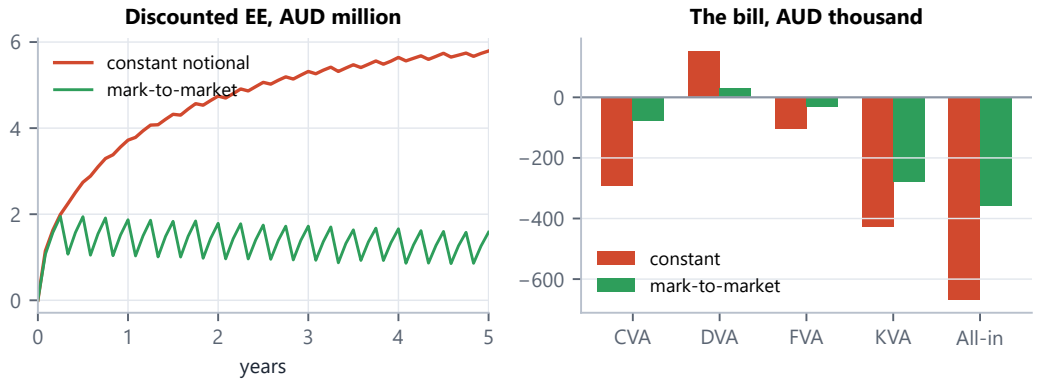



**Figure 14.1.** Left: discounted EE of the two swaps. Right: their bills, AUD thousand.

The constant-notional swap's exposure peaks at AUD 5.8 million; the resetting swap's is a sawtooth that never exceeds AUD 1.9 million, 3.0 times less. The market's resetting convention is, in effect, a quarterly collateral call on the principal.

## The bill

CVA, DVA and FVA come from the simulated exposure as in Chapters 6, 8 and 10. Capital follows Chapter 11 with SA-CCR's foreign-exchange add-on, 4% of notional (the interest-rate add-ons of the two floating legs are small and left out), the BBB risk weight and the reduced BA-CVA.

The full bill (negative = cost to us). FVA is the funding cost only; MVA is left out (see below).

| LINE | CONSTANT NOTIONAL (AUD) | MARK-TO-MARKET (AUD) |
|---|---|---|
| CVA | −290,711 | −78,168 |
| DVA | +152,924 | +30,350 |
| FVA (funding cost) | −104,122 | −30,886 |
| KVA | −428,024 | −278,312 |
| All-in | −669,932 | −357,016 |

The constant-notional bill is AUD 669,932, about 15.0 basis points a year; the mark-to-market bill AUD 357,016, about 8.0. On a desk that becomes a quote: Counterparty A would be asked for BBSW plus 30.5 basis points on a constant notional, or 23.5 with resets, against a par of 15.5. The resetting notional saves AUD 312,916 on one trade, which is the economic reason the AUD/USD market trades mark-to-market swaps.

Notice that KVA shrinks far less than CVA (1.5 times against 3.7). SA-CCR's add-on is a percentage of notional and doesn't see the resets: capital today is AUD 681,903 for both, and only the replacement cost benefits. When the rules don't see a risk reduction, it doesn't reach the price.

## What's left out

MVA: the margin rules exempt the fixed exchange of principal from initial margin, and here the interest-rate legs don't move, so little is left to margin. A full model adds a Hull-White factor for each currency, the three-factor model banks use. And if A's fortunes followed the Australian dollar, Chapter 13's wrong-way risk would enlarge every credit line.

## The series in five lessons

1. Exposure is an option: price it, and check the simulation against it.
2. XVA belongs to the netting set, not the trade.
3. Collateral turns credit risk into funding cost.
4. Count your own credit once.
5. For uncollateralised corporate trades, capital often costs more than credit.

**Bip's rule.** The risk-free price is the first line of the bill; XVA is the rest of it.

**Try this.** The final checkpoint is in the appendix.

---

#### Sources

Companion notebook: `s4e14_capstone_xccy.ipynb`. Every number in this chapter is computed by it.

1. Bank for International Settlements, *Basel Framework CRE52 (SA-CCR), Table 2*. https://www.bis.org/basel_framework/chapter/CRE/52.htm
2. Bank for International Settlements, *Basel Framework MGN (margin requirements for non-centrally cleared derivatives)*. https://www.bis.org/basel_framework/chapter/MGN/10.htm

CHAPTER 15 · VIDEO EPISODE 15

# How Banks Compute XVA

*Hundreds of thousands of trades, thousands of paths, every sensitivity, by the next morning.*

## The scale of the problem

Our running swap is one trade on 40,000 paths at about seventy dates: 2,880,000 valuations. A large bank's nightly run might be 500,000 trades on 5,000 paths at 100 dates (illustrative orders of magnitude): 250,000,000,000 valuations. At our speed of 0.13 microseconds per swap valuation that alone would take about 9.0 hours, before options, sensitivities or any trade that needs its own numerical method. Three ideas make it fit.

## Idea one: one world, simulated once

Every curve, exchange rate and credit spread the bank's trades depend on is simulated together on one set of paths, with correlations between them, and every trade is valued on those same paths. Trades with one counterparty can then be netted path by path (Chapter 7), and collateral applied (Chapter 9).

How many paths? Monte Carlo error falls with the square root of the number of paths:

The running swap's CVA on subsets of the same 40,000 antithetic paths.

| PATHS | CVA (AUD) | ± ONE SE (AUD) | SE (%) |
|---|---|---|---|
| 1,250 | 60,810 | 1,084 | 1.78 |
| 5,000 | 62,423 | 577 | 0.92 |
| 20,000 | 62,701 | 284 | 0.45 |
| 40,000 | 62,756 | 201 | 0.32 |

Thirty-two times the paths cut the error 5.4 times, close to $\sqrt{32}$. Halving the error costs four times the computing.

## Idea two: regression instead of pricing

Our swap has a formula for its value on any future date. Most trades don't: a Bermudan swaption on a future date needs its own model, and a simulation inside a simulation is hopeless at scale. **American Monte Carlo** estimates the value function by regression. On every path we know the cash flows the trade actually paid after date $t$; each path's discounted future cash flows are a noisy estimate of the value at $t$ given where the path was. Fit a smooth function of the state at $t$ through all of them, here a cubic in Hull-White's $x$, and that is the value function.

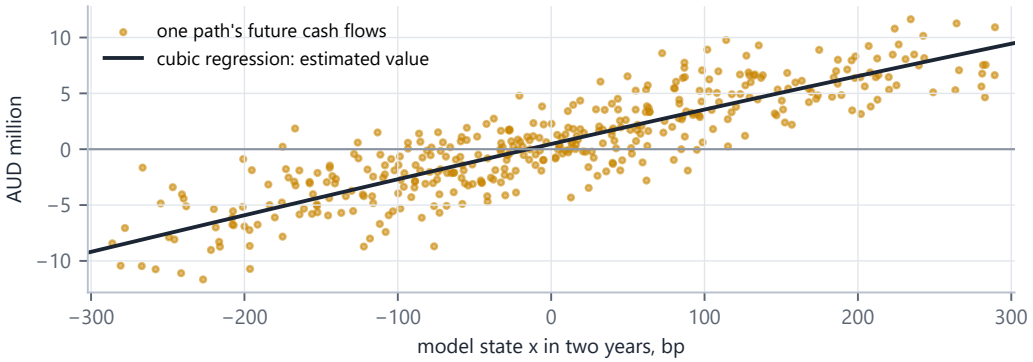



**Figure 15.1.** 400 paths' discounted future cash flows at year two against the model state, and the fitted value function.

Two years out, the expected exposure from regressed values is AUD 1,648,392 against AUD 1,644,988 from the exact formula: within 0.2%.

Run backwards through time, the same regression prices early exercise (Longstaff and Schwartz, 2001). For a payer Bermudan swaption on our swap, exercisable at 7 reset dates from year one, regression decides on each path whether to exercise:

| METHOD | PRICE (AUD) |
|---|---|
| Longstaff-Schwartz on our paths | 1,516,853 ± 5,444 |
| QuantLib finite differences | 1,513,457 |
| European, first date only | 932,901 |

They agree within 0.2%; the right to choose the date is worth AUD 580,556 more than a single European exercise.

Longstaff-Schwartz in the notebook.

```
for i in reversed(ex_i):
    k = int(np.argmin(np.abs(grid - sc.ts[i])))
    exv = sc.value_at(hw, sc.ts[i], X[:, k])
    itm = exv > 0
    if i == ex_i[-1]:
        cash = np.where(itm, D[:, k] * exv, 0.0)
        continue
    xk = X[itm, k]
    B_ = np.vstack([np.ones_like(xk), xk, xk**2, xk**3]).T
    cont = cash[itm] / D[itm, k]
    b = np.linalg.lstsq(B_, cont, rcond=None)[0]
    take = exv[itm] > B_ @ b
    idx = np.where(itm)[0][take]
    cash[idx] = D[idx, k] * exv[idx]
```

## Idea three: sensitivities

The XVA desk hedges, so it needs CVA's sensitivity to every credit spread and every rate quote, daily. Bump and revalue means one full run per input: 18 reruns for our one swap (six CDS tenors and the AONIA quotes). Most of the credit risk sits on the five-year CDS (AUD 363 per basis point of AUD 533), and the rate sensitivity is AUD 606 per basis point; the rate bumps alone took 5.8 seconds.

**Adjoint algorithmic differentiation** (AAD) records the valuation as it runs, then sweeps backwards through it once applying the chain rule, giving every sensitivity for a small constant multiple of one valuation's cost, however many inputs there are. It turned overnight XVA sensitivities from impossible into routine, at the price of rewriting pricing code.

## The rest of the toolkit

Paths are independent, so the work spreads across graphics processors and computing grids. The Open Source Risk Engine (ORE), built on QuantLib with Python bindings, is a full XVA engine to explore beyond these notebooks. During the day, an incremental quote for a new trade reuses last night's paths and netting-set values and prices only the new trade on them (Chapter 7). And often the hardest part is the data: every trade, netting agreement and CSA term, correct, every night.

More paths don't fix a wrong model, regression is checked against exact prices wherever possible, and sensitivities are usually the most expensive thing an XVA engine computes.

**Bip's rule.** Scale comes from clever maths first, and bigger computers second.

## Sources

Companion notebook: `s4e15_how_banks_compute.ipynb`. Every number in this chapter is computed by it.

1. SIAM, *Griewank, A. and Walther, A. (2008), Evaluating Derivatives: Principles and Techniques of Algorithmic Differentiation, 2nd ed., SIAM*. https://doi.org/10.1137/1.9780898717761
2. Oxford University Press, *Longstaff, F. A. and Schwartz, E. S. (2001), Valuing American Options by Simulation: A Simple Least-Squares Approach, Review of Financial Studies 14(1)*. https://doi.org/10.1093/rfs/14.1.113
3. Acadia / Open Source Risk Engine project, *Open Source Risk Engine (ORE)*. https://www.opensourcerisk.org/

# Final checkpoint and glossary

## Final checkpoint

Try these before looking at the answers.

1. A three-year swap has expected positive exposure of AUD 1.5 million in each year; the counterparty's default probabilities are 1%, 1.5% and 2% in years one to three; loss given default is 60%. Roughly what is the CVA?
2. Why doesn't daily variation margin with no threshold take exposure to zero?
3. Why shouldn't you count both DVA and the funding benefit (FBA)?
4. For the uncollateralised BBB corporate swap of this book, which is bigger, CVA or KVA?

**Answers.**

1. $0.6 \times 1.5\text{m} \times (1\% + 1.5\% + 2\%) = \text{AUD } 40{,}500$.
2. The margin period of risk: after the last margin call the counterparty meets, the trade keeps moving until it is closed out, and that move is uncollateralised (Chapter 9).
3. Both price your own credit on your negative exposure: a funding spread is largely a credit spread, so FBA and DVA measure the same thing twice (Chapter 10).
4. KVA: about 2.2 times CVA for the running swap (Chapter 11), and the largest line on both bills of Chapter 14.

## Glossary

**American Monte Carlo.** Estimating future values by regressing simulated future cash flows on the state, instead of pricing on every path.

**BA-CVA.** Basel's basic approach to capital for CVA risk; the reduced version uses sector risk weights, maturity and exposure at default.

**Bilateral CVA.** CVA minus DVA: the adjustment when both sides can default.

**Close-out netting.** Terminating all trades with a defaulted counterparty and adding them into one net amount.

**ColVA.** Collateral valuation adjustment: the value of collateral earning something other than the discounting rate, or of a choice of collateral.

**Credit triangle.** Hazard rate ≈ CDS spread ÷ (1 − recovery).

**CS01.** Change in a value for a one basis point rise in credit spreads.

**CSA.** Credit support annex: the collateral terms of a master agreement.

**CVA.** Credit valuation adjustment: the price of the counterparty's default.

**DVA.** Debit valuation adjustment: the value to us of our own default, priced on negative exposure.

**EE, EPE, effective EPE.** Expected exposure at a date; its time average; the regulators' running-maximum version over the first year.

**Euler allocation.** Splitting a netting set's CVA into trade shares that add up exactly.

**FCA, FBA, FVA.** Funding cost and benefit adjustments, and their net: exposure times funding spread times time.

**Hazard rate.** The instantaneous rate of default given survival so far.

**Hull-White.** A one-factor normal short-rate model fitted to today's curve.

**Incremental CVA.** The change in a netting set's CVA when a trade is added.

**Initial margin.** Collateral posted by both sides, segregated, covering the move over the margin period of risk.

**KVA.** Capital valuation adjustment: the cost of the return shareholders require on regulatory capital.

**Margin period of risk (MPoR).** The time from the last margin met to close-out; Basel floor 10 business days for daily-margined bilateral trades.

**Martingale test.** Checking that simulated discount factors average to today's curve.

**MVA.** Margin valuation adjustment: the cost of funding initial margin.

**PFE.** Potential future exposure: a high percentile (here 97.5%) of exposure at a date.

**Risk-neutral probability.** A probability implied from market prices, used for pricing, not forecasting.

**SA-CCR.** Basel's standardised approach for counterparty credit risk: EAD = 1.4 × (replacement cost + multiplier × add-on).

**SIMM.** ISDA's standard initial margin model, based on sensitivities, risk weights and correlations.

**Wrong-way risk.** Default likelier when exposure is high; right-way risk is the reverse.

**XVA.** The family of valuation adjustments between the risk-free price and the price actually charged.